O-C plug-in

Overview

This plug-in computes **O-C** (observed minus computed) values for times of light-curve extrema and displays an O-C diagram. A fixed test ephemeris (period and epoch) is compared against observed times of maximum (or minimum) light to reveal timing errors.

Primary reference for the six-clock O-C tutorial (Tables 13.1–13.2): AAVSO *Variable Star Astronomy*, chapter 13 — freely available PDF: Variable Stars and O–C Diagrams (Grant Foster). The same material appears in Foster, *Analyzing Light Curves*, ch. 13.

Two plug-ins are provided.

Plug-in	Menu	Role
O-C	**Tools** → **O-C…**	Main O-C tool (includes **Export CSV…** in results)
O-C demo data	**File** → **O-C demo data…**	Optional Foster clock light curves

The O-C tool can be opened from the **Tools** menu, even when no observations have been loaded, since other than a **From observations** option, an **Imported timings file** option allows a previously exported/created timings file to be opened instead.

Published help (PDF): O-C on aavso.github.io.

Installation

1. In VStar, choose **Tools** → **Plug-in Manager…**
2. Select and install **O-C** (required). Optionally install **O-C demo data** for Foster tutorial light curves.
3. **Close and restart VStar** so new menu items appear.

See also AAVSO VStar Plug-in Library.

Optional: **File** → **Preferences…** → **Plug-in Settings** to check the plug-in download URL matches your VStar version (see the VStar wiki installation recipes).

Quick start

1. **Tools** → **O-C…**
2. Set **data source**, ephemeris (period and epoch), and other options in the parameter dialog.
3. Click **OK** (a file chooser opens if you chose **Imported timings file**).
4. Review the results dialog: **O-C diagram**, **Data table**, and **Fit summary**.

See **Examples** for Foster demo walkthroughs. **Appendix A** covers all six clocks (demo light curves or imported timings). For a typical workflow on your own AID data (measure → export CSV → re-import), see **Appendix C**.

Examples

These three walkthroughs use the optional **O-C demo data** plug-in (no real star or downloaded timing file required). Install it if needed (see **Installation**), then restart VStar. Each example uses the VSA/Foster test ephemeris **P = 1.0 d**, **epoch = 2450000.0** (Tables 13.1–13.2 in the [Chapter 13 PDF]).

Example	Demo scenario (VSA clock)	Purpose
1	Foster clock 4 — epoch jump	Two-segment fit with **parallel** segments
2	Foster clock 5 — period change	Compare linear, quadratic, and two-segment (**different slopes**)
3	Foster clock 6 — slowing clock	Compare **linear** vs **quadratic** for a curved O-C

Clocks 1–3 (correct ephemeris, epoch offset, period error) are covered in **Appendix A**. Demo menu labels keep the “Foster clock” wording used in the plug-in; the numbers and tables are those of the VSA chapter.

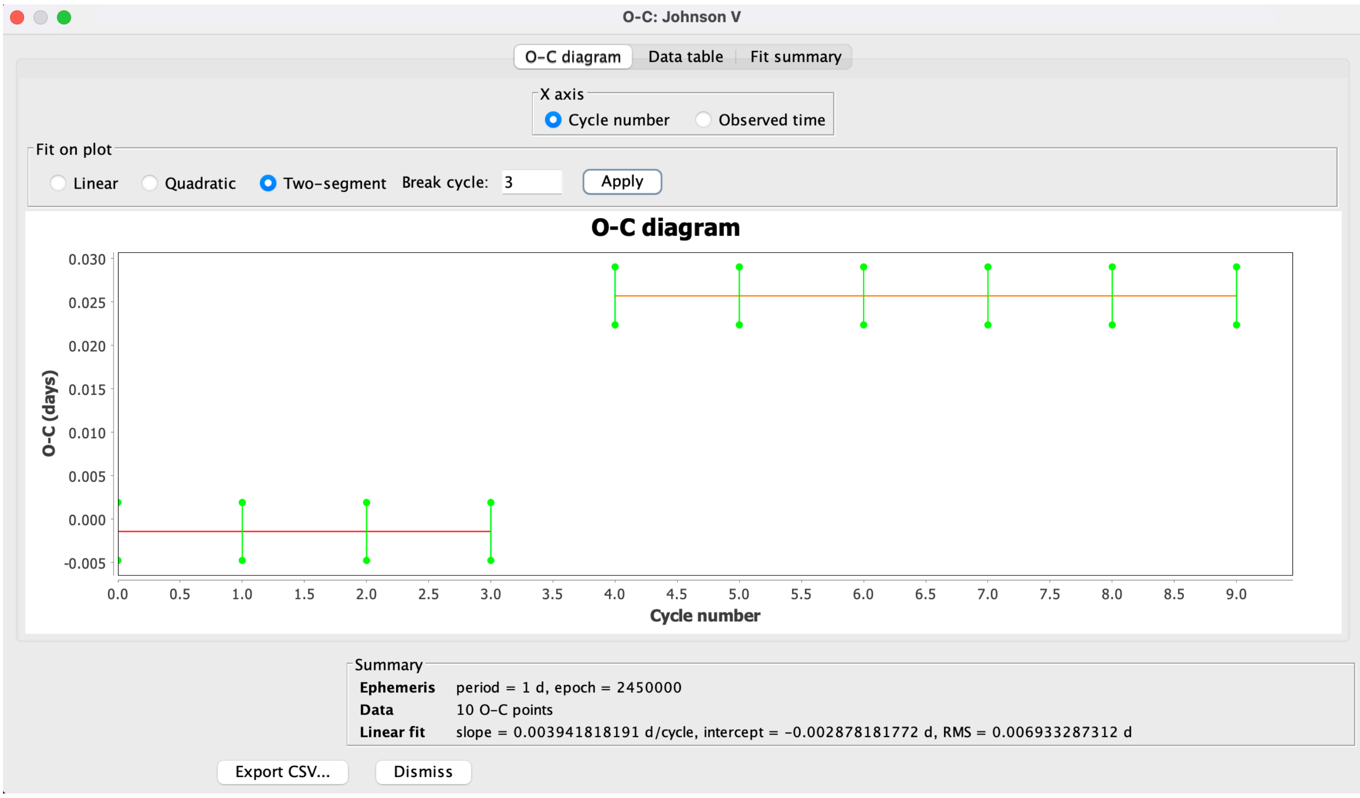
Shared setup for each example (details only in Example 1; abbreviated later):

1. **File** → **O-C demo data…** → choose the demo scenario for that clock → **OK**.
2. **Tools** → **O-C…** → **From observations** → **Star metadata** → **Event:** Maximum → **Timing method:** Parabolic interpolation → **OK** → select **Johnson V**.

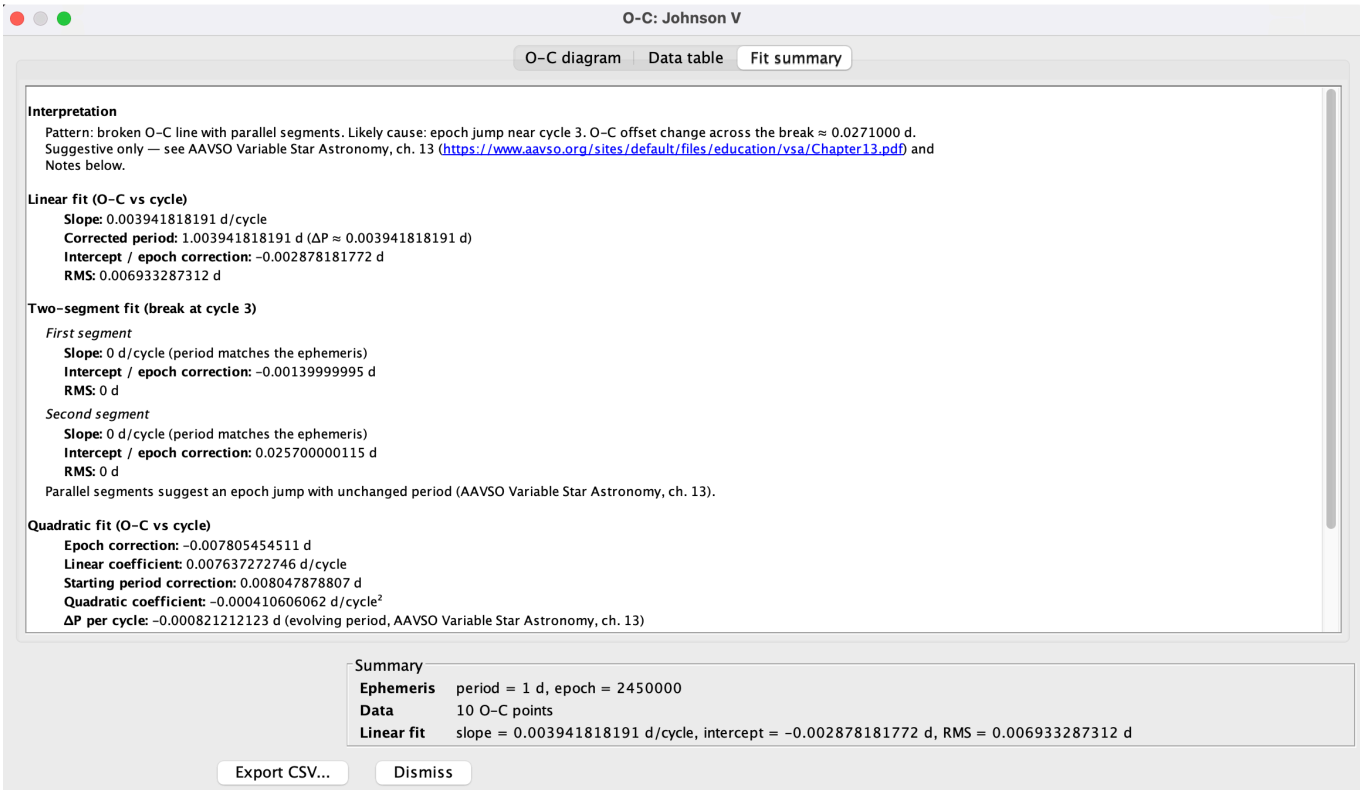
Example 1 — Epoch jump (clock 4)

A sudden step in O-C with unchanged slope on each side (VSA Table 13.2, clock 4).

1. Load **Foster clock 4 — epoch jump** and open O-C as in the shared setup. Note the load-message hint: break cycle **3** (cycles 0–3 are the pre-jump plateau; cycle 4 is already post-jump).
2. On the **O-C diagram**, the points step upward at cycle 4. Select **Two-segment**, enter **Break cycle 3**, click **Apply**. Both segments should be nearly flat (slope ≈ 0) with different intercepts.



3. Open **Fit summary**. The **Interpretation** should describe parallel segments (epoch jump with unchanged period).

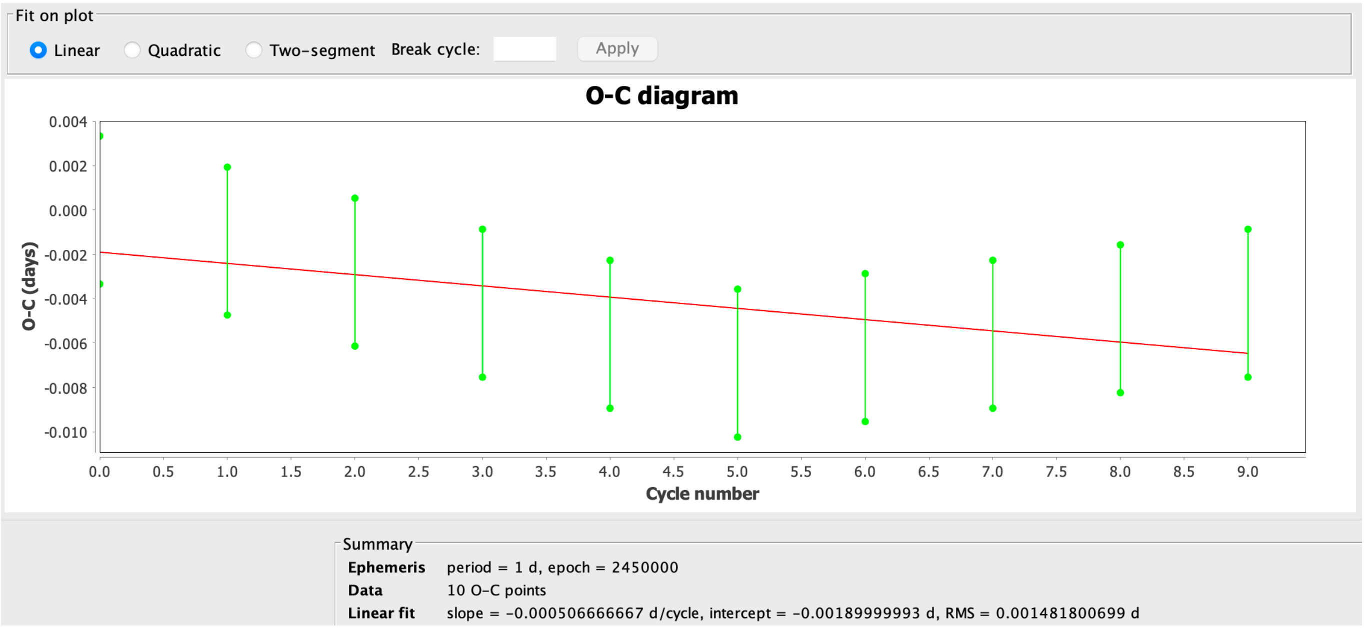


Example 2 — Period change (clock 5)

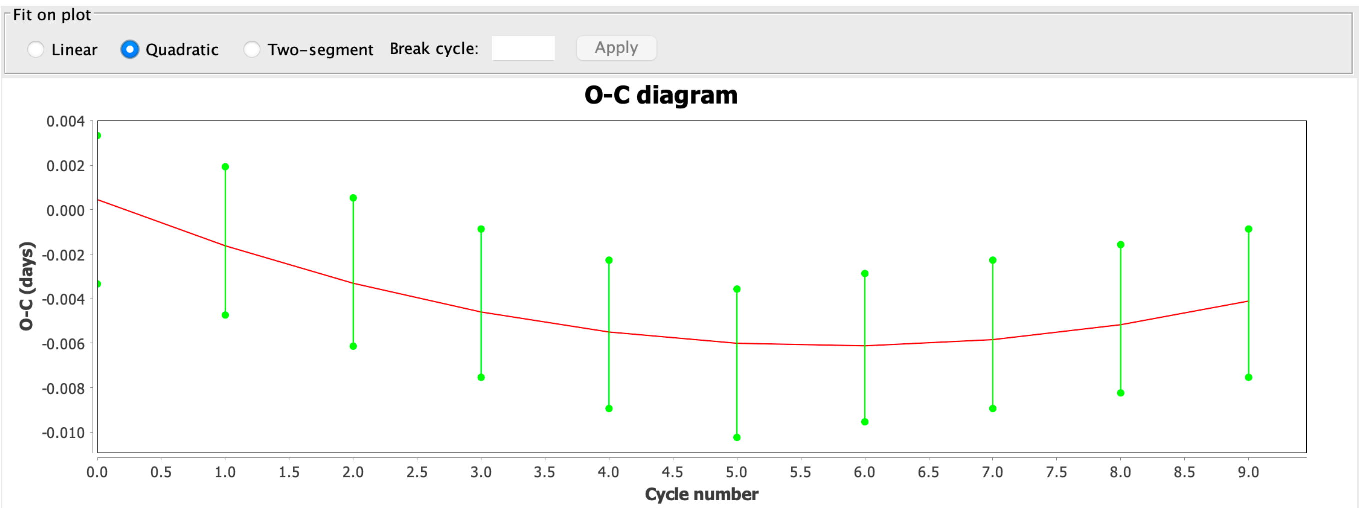
A slope change mid-run: a single line or parabola is a poor model; two segments with different slopes match

Table 13.2 (clock 5).

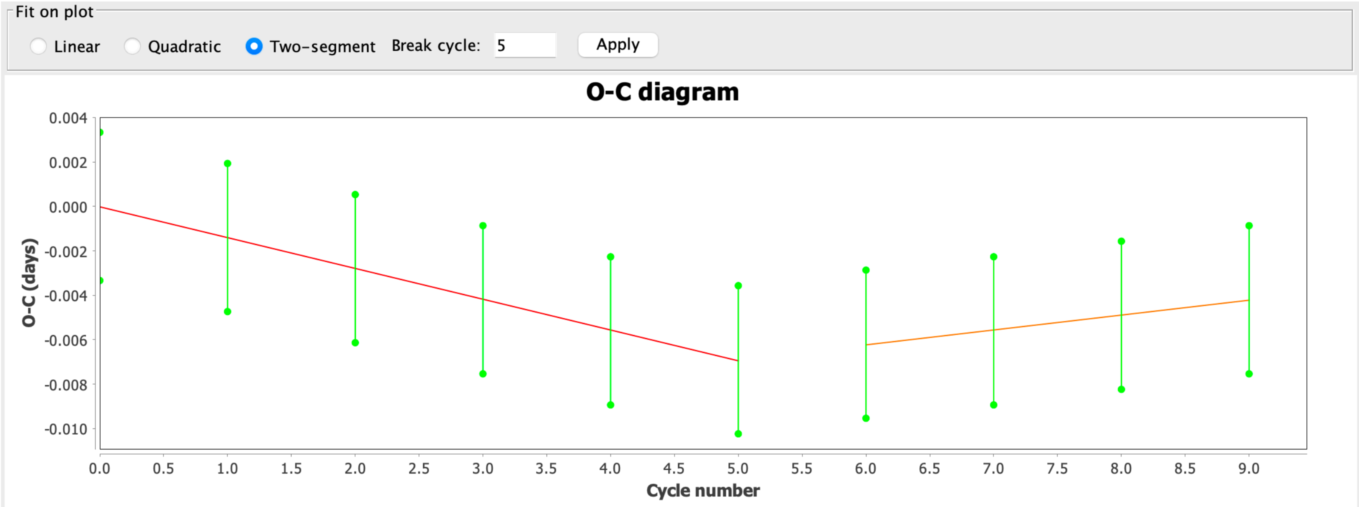
1. Load **Foster clock 5 — period change** and open O-C as in the shared setup. Note the hint: break cycle **5** (cycles 0–5 are the first slope; cycle 6 starts the new regime).
2. Leave **Fit on plot** on **Linear**. The points show a kink; one line is only a compromise across both segments.



3. Switch to **Quadratic**. A parabola may look somewhat better, but it still does not capture a sharp period change at one cycle.



4. Switch to **Two-segment**, enter **Break cycle 5**, click **Apply**. Separate lines before and after the break should show different slopes.



5. Open **Fit summary**. With **Two-segment** selected, the **Interpretation** should report a period change. Switch **Fit on plot** back to **Linear** or **Quadratic** and return to **Fit summary** — the text follows the fit shown on the diagram. Compare with Example 1 (parallel segments / epoch jump).

Interpretation

Pattern: broken O–C line with different slopes. Likely cause: period change near cycle 5. Segment period change ≈ 0.0020557 d/cycle (second minus first segment). Suggestive only — see AAVSO Variable Star Astronomy, ch. 13 (https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf) and Notes below.

Linear fit (O–C vs cycle)

Slope: −0.000506666667 d/cycle
Corrected period: 0.999493333333 d (ΔP ≈ −0.000506666667 d)
Intercept / epoch correction: −0.00189999993 d
RMS: 0.001481800699 d

Two-segment fit (break at cycle 5)

First segment

Slope: −0.00138571426 d/cycle
Corrected period: 0.99861428574 d (ΔP ≈ −0.00138571426 d)
Intercept / epoch correction: −0.000019047587 d
RMS: 0.000028171761 d

Second segment

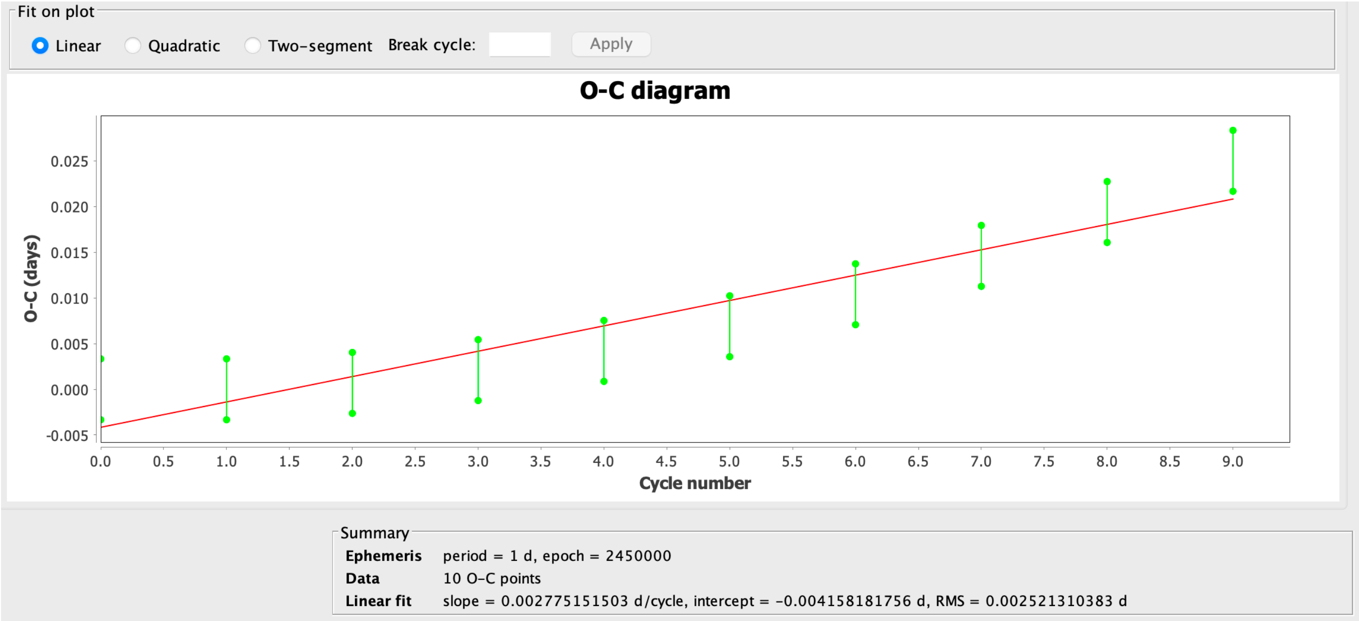
Slope: 0.000670000073 d/cycle
Corrected period: 1.000670000073 d (ΔP ≈ 0.000670000073 d)
Intercept / epoch correction: −0.010250000516 d
RMS: 0.000027386039 d

Different slopes suggest a period change near cycle 5 (AAVSO Variable Star Astronomy, ch. 13).

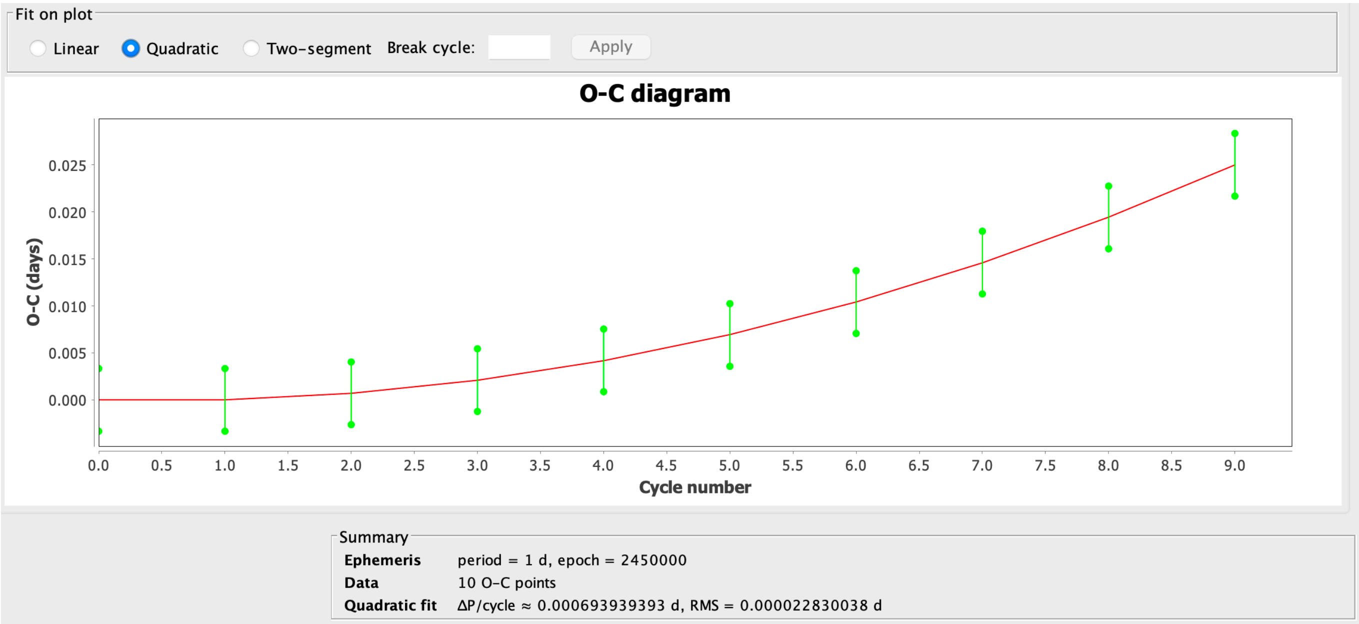
Example 3 — Evolving period (clock 6)

A smoothly curving O-C from an evolving period (Table 13.2, clock 6). Here **linear** vs **quadratic** is the main contrast (no break cycle required).

1. Load **Foster clock 6 — slowing clock** and open O-C as in the shared setup.
2. Leave **Fit on plot** on **Linear**. A straight line cannot follow the curve.



3. Switch to **Quadratic**. The parabola should track the points.



4. **Fit summary** should interpret a curved O-C (evolving period) when **Quadratic** is selected.

Interpretation

Pattern: curved (parabolic) O–C. Likely cause: evolving period. ΔP/cycle ≈ 0.0006939 d. Suggestive only — see AAVSO Variable Star Astronomy, ch. 13 (https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf) and Notes below.

Linear fit (O–C vs cycle)

Slope: 0.002775151503 d/cycle
Corrected period: 1.002775151503 d (ΔP ≈ 0.002775151503 d)
Intercept / epoch correction: −0.004158181756 d
RMS: 0.002521310383 d

Quadratic fit (O–C vs cycle)

Epoch correction: 0.000005454604 d
Linear coefficient: −0.000347575768 d/cycle
Starting period correction: −0.000694545464 d
Quadratic coefficient: 0.000346969697 d/cycle^2
ΔP per cycle: 0.000693939393 d (evolving period, AAVSO Variable Star Astronomy, ch. 13)
RMS: 0.000022830038 d

Notes

Caution (AAVSO Variable Star Astronomy, ch. 13): for long-period variables and other stars with cycle-to-cycle period scatter, O–C residuals are not white noise — apparent trends may reflect intrinsic period jitter rather than a true ephemeris change. See also Foster (1993, JAAVSO 22, 145) on O–C autocorrelation. Chapter PDF: https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf

O–C pattern reference: AAVSO Variable Star Astronomy, ch. 13

Further reading — X Tri, Z Tau, and other real stars

The VSA chapter discusses real variables (e.g. **X Tri**, **Z Tau**, SU Vir) with O-C diagrams built from published **times of extremum**, not re-measured light curves. Bundled fixtures:

Star	Role	File
X Tri (EB)	VSA Table 13.9 minima	`xtri_table_13_9.txt`
Z Tau (Mira)	AAVSO LPV times of **maximum**	`ztau_maxima.txt`

Downloads: xtri*table*13_9.txt, ztau_maxima.txt.

Reproduce Activity 13.6 (X Tri):

1. Optionally load **X Tri** from the AAVSO International Database with a JD range covering Table 13.9 minima (≈ **2442720 – 2450090**), Visual or any band you prefer. The light curve is context only (e.g. for vertical markers); the O-C points come from the imported timings file, not from re-timing this sparse photometry.
2. **Tools** → **O-C…** → **Imported timings file** → choose `xtri_table_13_9.txt`.
3. Period **0.975352** d and epoch **2442502.721** fill from the file's `# period=…, epoch=…` comment when present; set **Event** to **Minimum**.
4. Inspect O-C vs cycle (and try **Quadratic** on Fit on plot). Expect a clear long-term trend — the teaching linear ephemeris is **not** a perfect flat O-C; that is the point of the exercise.

Z Tau (long-period / Mira maximum O–C):

1. Optionally load **Z Tau** from AID with **JD ≈ 2417000 – 2458100** (roughly 1905–2018, spanning the bundled maxima). Visual is fine for context; sparse Mira data are not meant for auto-timing this

exercise.

2. **Tools** → **O-C…** → **Imported timings** → `ztau_maxima.txt` (source: [AAVSO Maxima & Minima of LPVs]).
3. Period **466.2** d and epoch from the file header fill automatically; set **Event** to **Maximum**.
4. Expect a **curved / multi-segment** O–C: Z Tau's period has evolved (historical means ~500 d early, GCVS later ~466 d, VSX often ~446 d). A fixed trial **P** is deliberately imperfect—the diagram is the lesson.

Do **not** expect to match these diagrams by auto-timing sparse AID Visual photometry: they use **published times of extremum**. For other stars, use literature timings or measure dense photometry, then **From observations** / **Edit timings**.

Clock demos (Tables 13.1–13.2) remain in **Examples** / **Appendix A**. For an AID export/re-import outline on real photometry, see **Appendix C**.

For **eclipsing binaries**, decades of **eclipse timings** can reveal **apsidal motion** (sinusoidal O-C from general relativity and tidal distortion) or **period change** from mass transfer; those effects need long baselines.

Results dialog

After O-C is computed, a modeless results window opens with three tabs:

Tab	Contents
O-C diagram	O-C points (optional error bars); **X axis** and **Fit on plot** controls
Data table	Cycle, observed time, computed time, O-C, uncertainty, obs count, QC
Fit summary	Pattern interpretation for the selected fit, plus fit details

Fit on plot — radio buttons choose which fit is drawn:

Option	When available	Notes
Linear	≥2 O-C points	Default; single least-squares line
Quadratic	≥3 O-C points	Curved trend (Foster clock 6 / evolving period)
Two-segment	≥4 O-C points (≥2 per segment)	Enable **Break cycle** + **Apply** on the same row

Fit summary — leads with an **Interpretation** paragraph for the fit currently selected on the diagram (VSA / Foster ch. 13 patterns). Fit coefficients and the LPV period-scatter caution follow below.

Light-curve check — solid vertical **O** markers (observed timing) and lighter dashed **C** markers (ephemeris prediction) are drawn on the raw-data light curve. Re-running O-C replaces previous O-C markers; loading a new star clears them. Labels (cycle numbers) appear only when there are ≤ 30 markers.

Edit timings — at the top of the results dialog:

Control	Action
Place O on light curve	Click the raw LC to place a free JD as an observed time. Chart pan/zoom drag is disabled while this is on.
Snap to nearest observation	When placing or dragging, snap O to a nearby observation (when a series is available).
Remove selected	Delete the selected table row / O marker (**Delete** key).
Clear all	Remove every timing.
Drag **O** marker	Press near an O vertical and drag to a new free JD.
Data table **O (time)**	Editable; type a new JD and the O-C list rebuilds.

C markers stay ephemeris-driven (not editable). QC notes show `manual`, `snap`, or the automatic method (e.g. KvW). Export CSV reflects the current edited list.

For **Kwee–van Woerden** seeds, the summary still reports how many cycles were timed versus skipped before editing; per-point **σ(O-C)** / **QC** may change when you replace automations with free JDs.

Export CSV… on the results dialog saves O-C points and fit metadata to a file.

Components (summary)

O-C tool

Parameter dialog uses a compact two-column layout. **Ephemeris source** is a drop-down only (not typable); edit **Period** and **Epoch** in their fields.

Data source options:

- **From observations** — automatic timing methods on a selected series.
- **Imported timings file** — load O times from a file.
- **Edit timings on light curve** — open an empty editor (period/epoch required) and place/remove O markers interactively. Optionally pick a series for snap.

After **OK**, choose an observation series when needed; the results dialog opens next. Use **Export CSV…** there to save results.

O-C demo data (observation source)

Optional. Loads synthetic light curves whose **maximum timings** match Tables 13.1–13.2 of the VSA Chapter 13 PDF (clocks 1–6). Bump shape is identical for every clock; only timing differs. Then run **Tools → O-C…** with **From observations** and the suggested ephemeris from the load message (P = 1 d).

Appendix A — Validation against VSA Tables 13.1–13.2

The six test clocks (theory **P = 1 day**, **epoch = cycle 0**) are from AAVSO *Variable Star Astronomy* chapter 13 (PDF). Computed time C*n = n. O-C = O*n – C_n (Table 13.2). The same tables appear in Foster, *Analyzing Light Curves*, ch. 13.

Reference O-C values (Table 13.2)

Cycle	Clock 1	Clock 2	Clock 3	Clock 4	Clock 5	Clock 6
0	0	+0.0035	0	−0.0014	0	0
1	0	+0.0035	+0.0021	−0.0014	−0.0014	0
2	0	+0.0035	+0.0042	−0.0014	−0.0028	+0.0007
3	0	+0.0035	+0.0062	−0.0014	−0.0042	+0.0021
4	0	+0.0035	+0.0083	**+0.0257**	−0.0056	+0.0042
5	0	+0.0035	+0.0104	+0.0257	−0.0069	+0.0069
6	0	+0.0035	+0.0125	+0.0257	−0.0062	+0.0104
7	0	+0.0035	+0.0146	+0.0257	−0.0056	+0.0146
8	0	+0.0035	+0.0167	+0.0257	−0.0049	+0.0194
9	0	+0.0035	+0.0188	+0.0257	−0.0042	+0.0250

Demo scenario mapping

VSA / Foster clock	Demo menu label	Expected in plug-in
1	Foster clock 1 — correct ephemeris	Flat O-C ≈ 0
2	Foster clock 2 — epoch offset	Flat O-C ≈ +0.0035 d
3	Foster clock 3 — period error	Slope ≈ +0.0021 d/cycle
4	Foster clock 4 — epoch jump	Step at cycle 4; break **3**
5	Foster clock 5 — period change	Slope change; break **5**
6	Foster clock 6 — slowing clock	Curved O-C; select **Quadratic** on Fit on plot

Method A — imported timings

For each clock, download the timing file (links below), then:

1. **Tools** → **O-C…** → **Imported timings file**.
2. **Period:** 1.0 d, **Epoch:** 2450000.0.
3. Compare O-C and **Fit summary** to Table 13.2.

Method B — demo observation source

See **Examples** for clocks 4–6. For any clock:

1. **File** → **O-C demo data…** → pick a Foster clock (labels match the demo scenario mapping table above).
2. **Tools** → **O-C…** → **From observations**; use the suggested ephemeris from the load message (**Star metadata** pre-fills period and epoch).
3. Compare O-C, **Fit on plot**, and **Fit summary** to Table 13.2.

Timing files (Table 13.1 → times)

Epoch **2450000.0**, period **1.0** d for O-C tool entry. Download from aavso.github.io (published with plug-in docs; same files in `plugin/doc/foster/` in the repo).

Clock	Scenario	Download
1	Correct ephemeris	foster_clock1.txt
2	Epoch offset	foster_clock2.txt
3	Period error	foster_clock3.txt
4	Epoch jump (try break cycle 3)	foster_clock4.txt
5	Period change (try break cycle 5)	foster_clock5.txt
6	Slowing clock (Quadratic on Fit on plot)	foster_clock6.txt

Timing files — published extrema (X Tri, Z Tau)

File	Contents	Ephemeris
xtri*table*13_9.txt	122 published minima (cycle + JD); repeated cycles = different observers	(P = 0.975352\,\mathrm{d}), (t_0 = 2442502.721)
ztau_maxima.txt	84 AAVSO LPV times of **maximum** (cycle + JD)	(P = 466.2\,\mathrm{d}), (t_0 = 2415246.2)

Star	Optional AID JD load (context LC)	Event
X Tri	**≈ 2442720 – 2450090**	Minimum
Z Tau	**≈ 2417000 – 2458100**	Maximum

See **Further reading** above.

Appendix B — Reference

Data source in the parameter dialog is either **From observations** or **Imported timings file**. Period and epoch define the test ephemeris (required; same time system as the data — see below).

Ephemeris source

When **From observations** is selected:

UI label	Meaning
Phase plot	Period and epoch from the **active phase plot** (View → Phase Plot, or Period Analysis → New Phase Plot)
Star metadata	Period and epoch from the **loaded star record** (often from an AID download or other source that supplies catalogue fields)
Manual entry	Type period and epoch yourself

Disabled for **Imported timings** — enter period and epoch directly, unless the file is an O-C export CSV (metadata is read from the file).

Imported timings file format

One timing per line: `cycle time [sigma]` or `time [sigma]`. Whitespace, comma, or semicolon separators. `#` comments allowed.

Use **one consistent time system** for observations, epoch, and imported timings (JD, HJD, BJD_TDB, etc.). The plug-in does not convert between systems; mix them and O-C will be wrong. Homogenise with VStar's HJD/BJD tools first if needed.

You can also re-open a file saved with **Export CSV…** from a previous O-C run. The tool reads timings from the data rows and, for export files, fills in **Period**, **Epoch**, and **Event** from the `# period=…, epoch=…` comment and the `Event` column (leave those fields blank in the parameter dialog). Export columns are `O_time` and `C_time` (not a specific JD flavour).

Timing methods (From observations)

Observations only (with **Extreme N%**, **KvW folds**, and **Minimum observations per cycle** as applicable):

- **Parabolic interpolation** — parabolic refinement at discrete extremum (default; good for smooth peaks such as the Foster demos).
- **Mean JD of extreme N%** — mean time of brightest (or faintest) N% in each cycle.
- **From current model function** — analytic extrema from a JD-based model.
- **Kwee–van Woerden** — single-eclipse time-of-minimum/maximum (Deeg 2020), via the same library as **Tools** → **Kwee–van Woerden…**. After cycle bucketing, O-C trims an eclipse-like window around the discrete extremum (half-width capped at ~0.2 *P*), requires at least **7** points in that window, then runs KvW. Failed or undersampled cycles are **skipped** (no silent fall-back to parabolic). Use **Minimum** for eclipsing binaries, a good trial period / epoch so one eclipse sits per cycle bin, and inspect the vertical markers. Recommended min observations per cycle: **≥ 7**.

Event type

- **Maximum** or **Minimum**
- **Both maxima and minima** — for eclipsing binaries on real data (not used in six-clock tutorial in the VSA Chapter 13 PDF)

Interpreting O-C diagrams

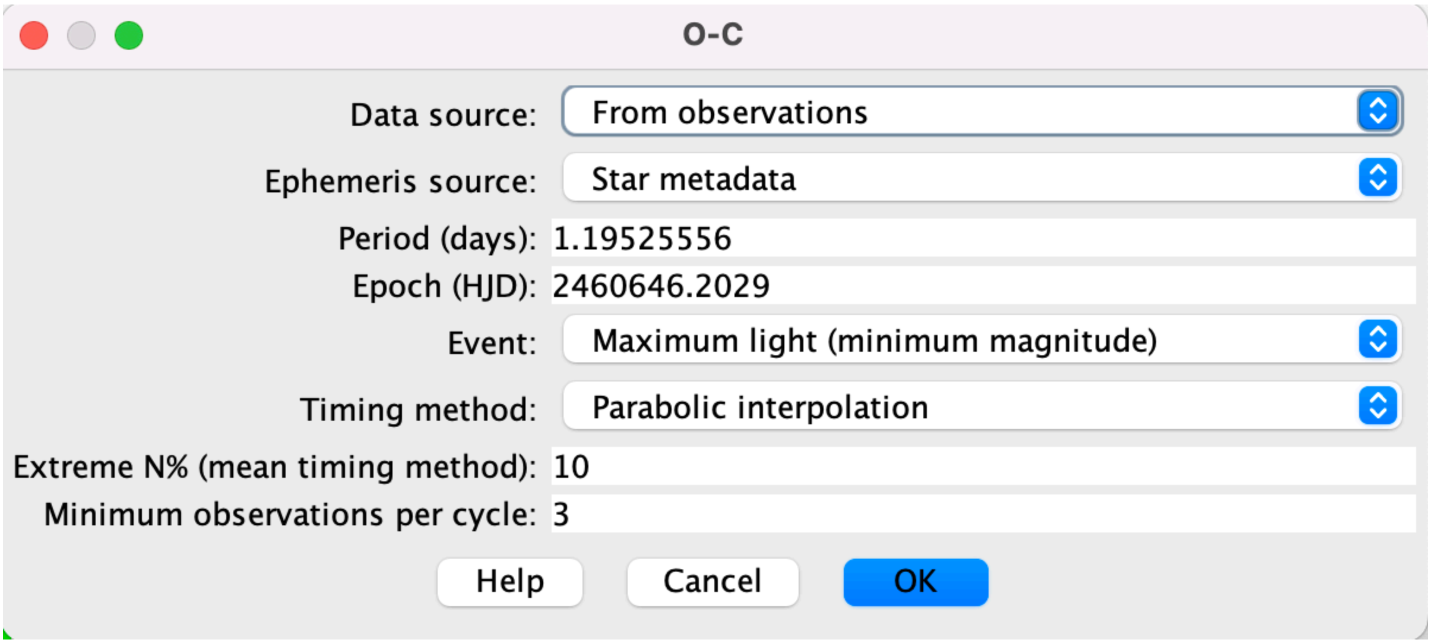
Pattern	Likely cause
Flat at 0	Ephemeris matches
Flat, offset	Epoch wrong, period OK
Linear slope	Period wrong
Broken line, parallel segments	Epoch jump
Broken line, different slopes	Period change
Curved (parabolic)	Evolving period → select **Quadratic** on the diagram

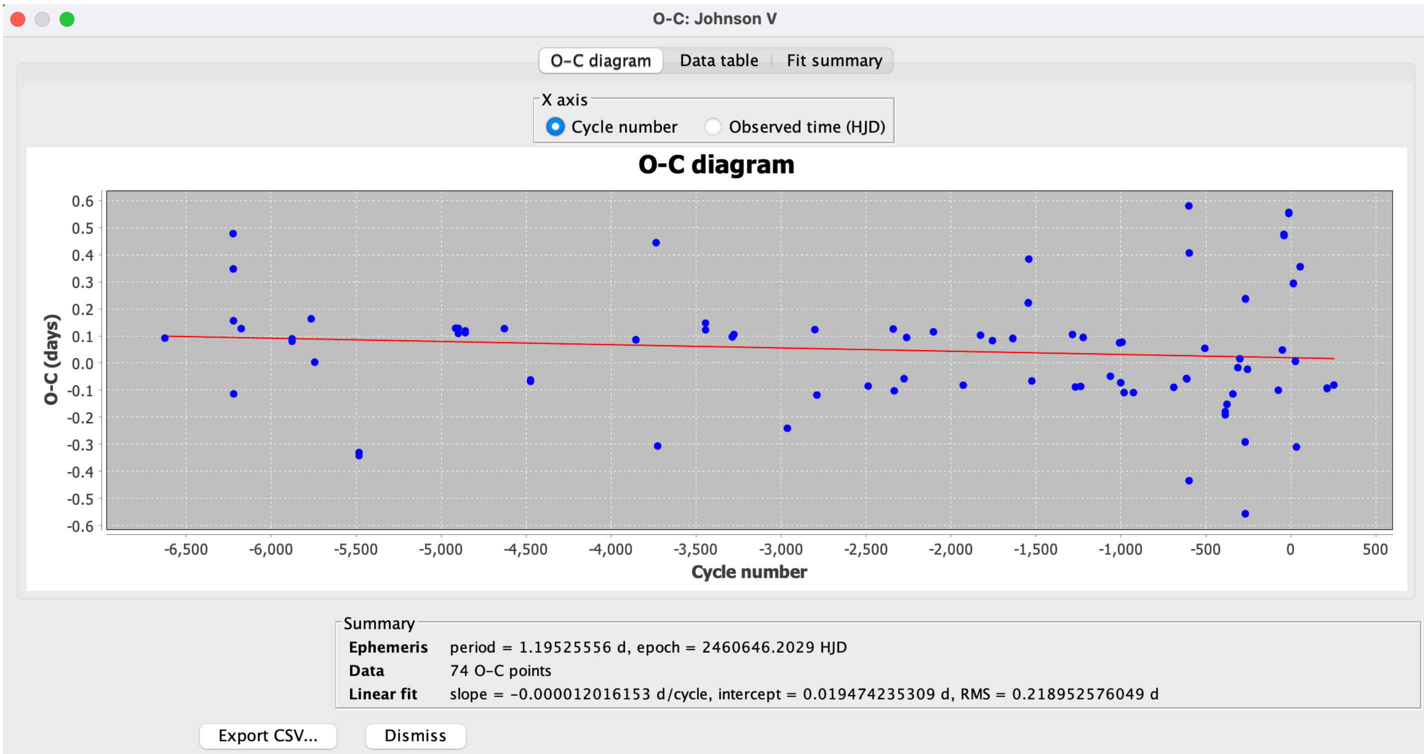
Appendix C — AID workflow (export and re-import)

Typical path on your own data: measure extrema from a loaded light curve, export timings, then re-open them without reprocessing the AID.

Suggested target: RZ Cas (AUID `000-BBF-490`), Algol-type EA. Period ≈ **1.20 d** (VSX). **Suggested JD range: 2452700 – 2461000** (about 22 years); load **V** only.

1. Load **RZ Cas** from the AAVSO International Database with the JD range and **V** series above.
2. **Tools** → **O-C…** → **From observations** → **Star metadata** → **Event:** Minimum → **OK** → select **V**.
3. Review the diagram, then **Export CSV…** (e.g. `rz_cas_timings.csv`).
4. **Tools** → **O-C…** again → **Imported timings file** → select the CSV. Period, epoch, and event are taken from the export metadata when present.





References

- AAVSO *Variable Star Astronomy*, chapter 13 (Grant Foster) — free PDF: [Variable Stars and O–C Diagrams] (primary reference for Tables 13.1–13.2 and the six-clock examples)
- Foster, G. *Analyzing Light Curves*, O-C Analysis, ch. 13, pp. 263–282 (same clock material; book form)
- [VStar plug-in library]
- [VStar issue #93]
- Deeg, H. J. (2020). The Kwee–van Woerden method revisited. *Research Notes of the AAS* / arXiv:2011.09231 — used by the KvW plug-in and O-C KvW path

- Kwee–van Woerden plug-in (standalone tool; S(T) / mirror QC remains there)

Version history

- **1.13** — Integrated **Edit timings** on results dialog: free-JD place mode (optional snap-to-obs), drag O markers, remove/clear, editable O column; **Data source: Edit timings on light curve** for empty start. Model rebuilds O-C / fits / markers live from the edited list. Bundled VSA **Table 13.9** X Tri minima (`plugin/doc/foster/xtri_table_13_9.txt`) and AAVSO **Z Tau** maxima (`plugin/doc/foster/ztau_maxima.txt`) for published-extrema exercises.
- **1.12** — **Kwee–van Woerden** timing method for from-observations O-C (eclipse windowing; σ_Deeg; skip undersampled cycles; packaging depends on KweeVanWoerdenLib). Light-curve check uses vertical **O/C** domain markers instead of the synthetic **O-C extrema** series. Data table **QC** column and KvW timed/skipped summary.
- **1.11** — From-observations runs add an **O-C extrema** series on the light curve (markers at measured O, mag) for checking timings; re-run replaces the series. Imported timings do not.
- **1.10** — Examples rewritten around VSA/Foster clocks 4–6; cite free VSA Chapter 13 PDF as primary reference for Tables 13.1–13.2; AID export/re-import moved to Appendix C; Results dialog docs updated for **Fit on plot** and interpretation.
- **1.9** — `OCAnalysisTool.test()` smoke check (Foster clock 2) for plug-in harness coverage.
- **1.8** — Neutral time labels (`O_time` / `C_time`, Epoch, Observed time); docs stress a consistent time system (JD/HJD/BJD/…) rather than assuming HJD. Removed unused O-C CSV observation-sink plug-in (export is results-dialog only).
- **1.7** — Added screenshots for parameter dialog, results diagram, Fit summary, and two-segment fit.
- **1.6** — User-facing rename: **O-C** (Tools menu) and **O-C demo data** (File menu); plug-in group **Timing**.
- **1.5** — Example 2 uses O-C demo data (Foster clock 2); Foster `.txt` downloads remain in Appendix A Method A only.
- **1.4** — User-facing doc restructure: Plug-in Manager install, examples, appendices, Foster timings on aavso.github.io (absolute URLs), screenshot placeholders.
- **1.3** — Foster Table 13.1 demo timings; six clocks only; compact dialog; general tool; break cycle on Fit summary.
- **1.2** — Imported timings, CSV export, quadratic fit, BOTH event type, demo observation source.
- **1.1** — Model timing, linear/two-segment fits, error bars.
- **1.0** — Initial observation-based O-C plot and table.